DH

DeliveryHub

ADMIN PORTAL

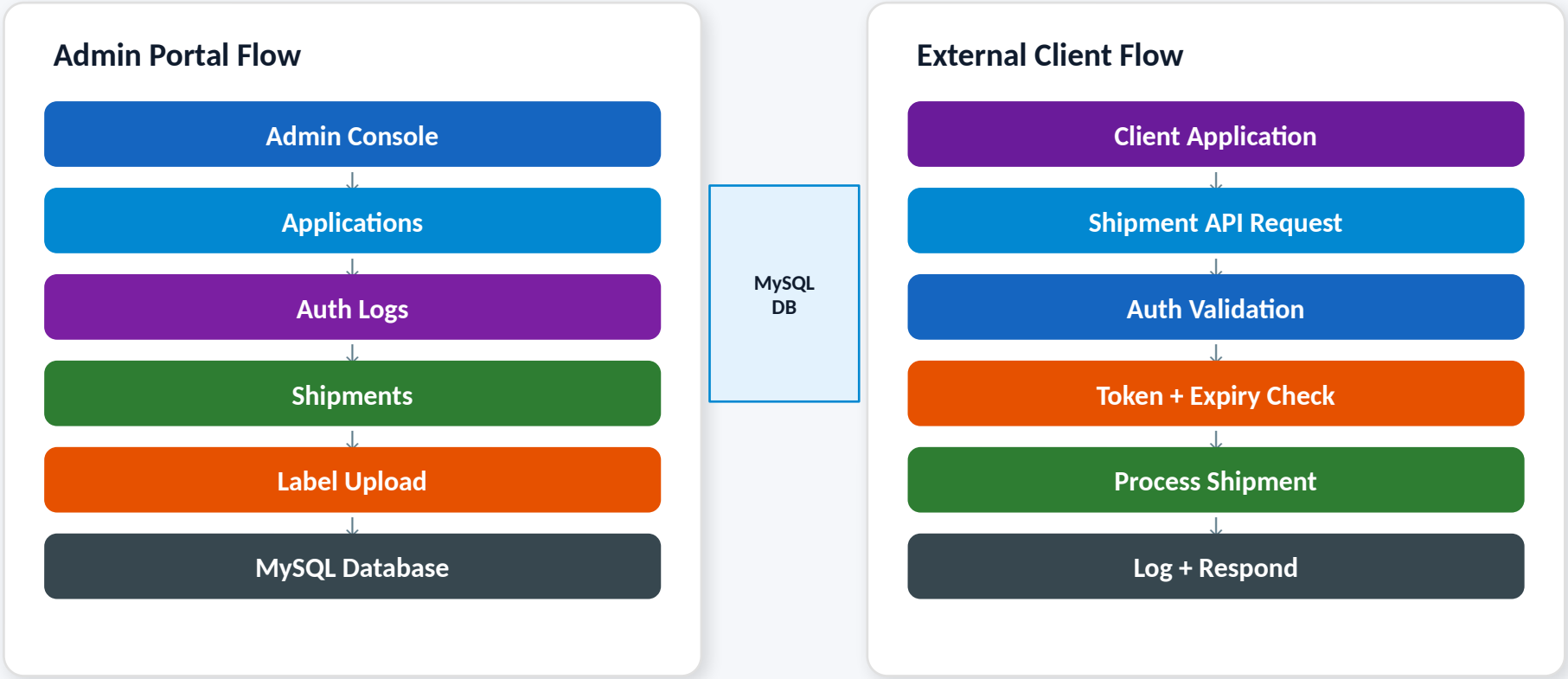
Secure Shipment Management & Authentication System

A Flask-based platform for secure application onboarding, shipment lifecycle management, and audit-ready authentication logging.

Flask | MySQL | Python | REST API

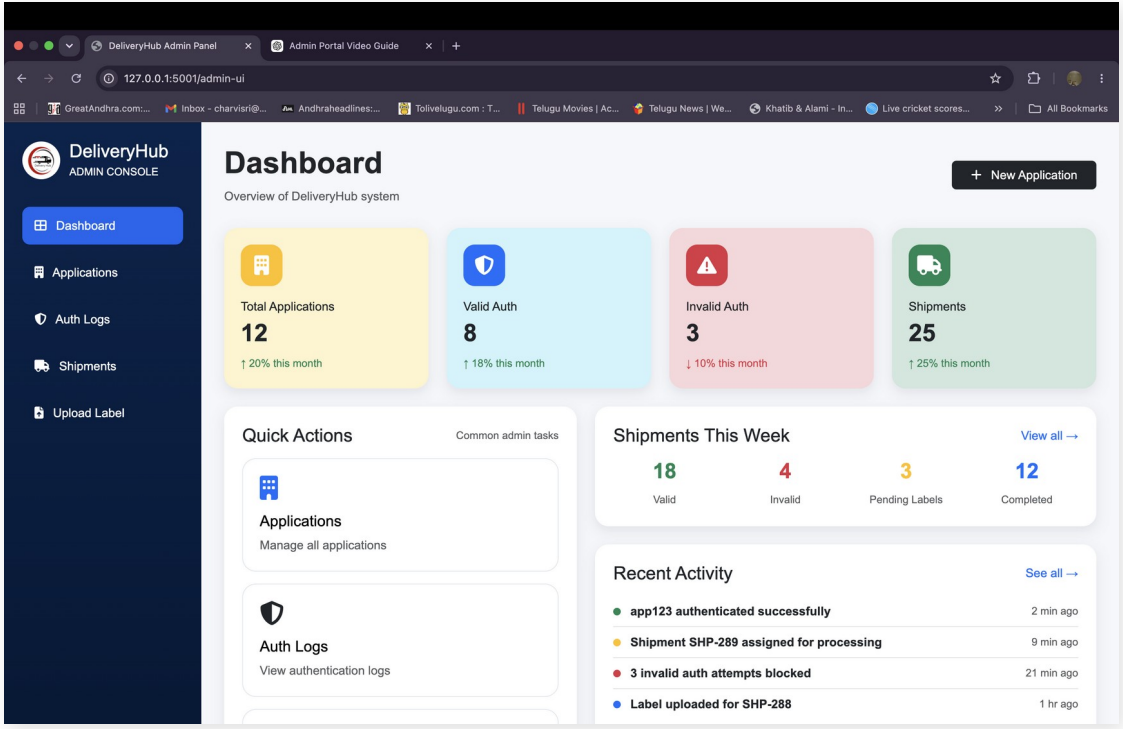
System Architecture

How DeliveryHub components interact



Admin Dashboard

System-wide overview at a glance



Total Applications

All onboarded client apps

Valid Auth

Successful auth attempts

Invalid Auth

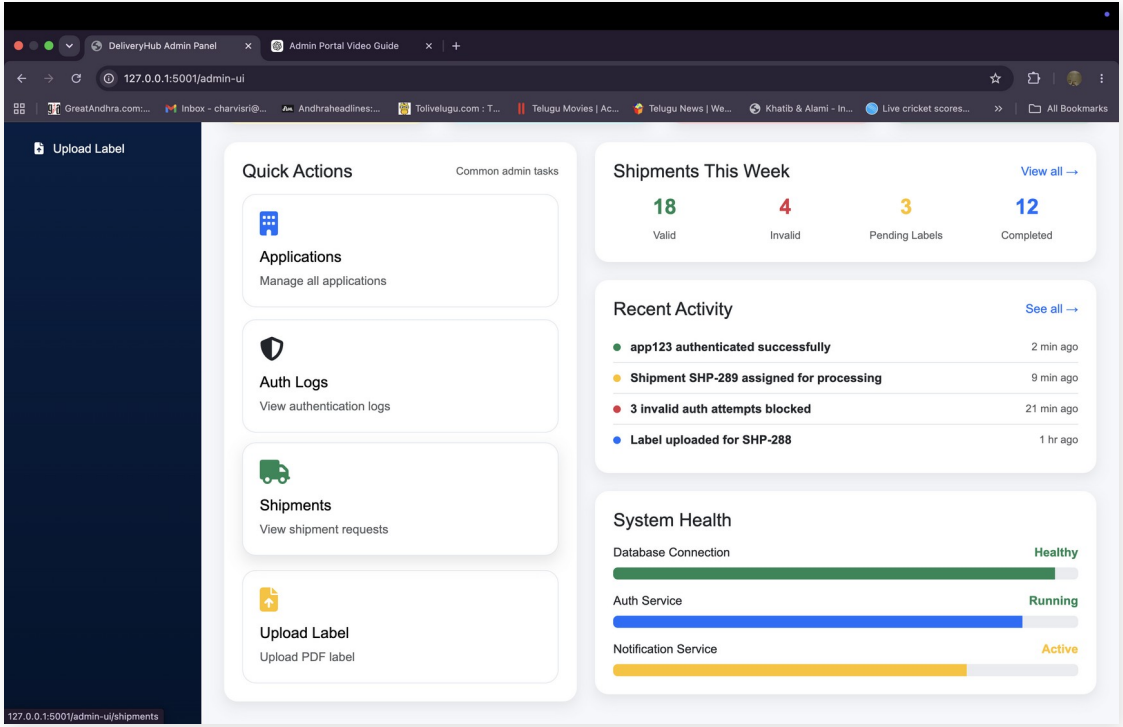
Failed / blocked requests

Shipments

Total requests processed

Dashboard — Quick Actions & Health

Shortcuts and live service monitoring



System Health

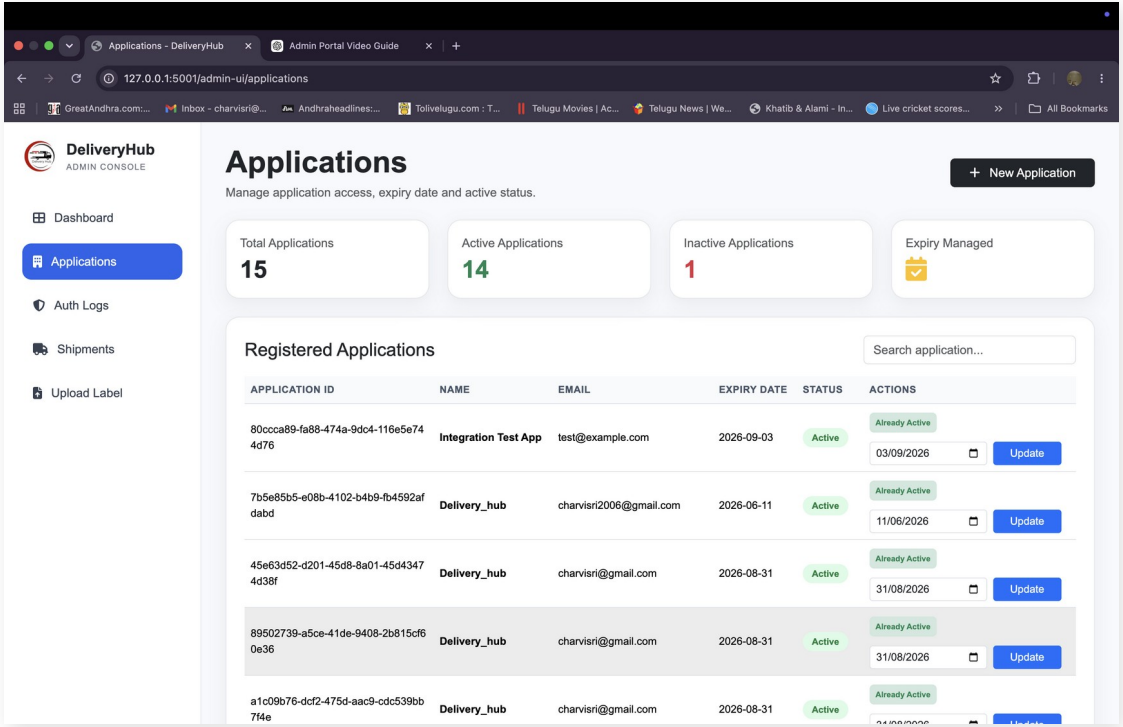
Database Connection — Healthy

Auth Service — Running

Notification Service — Active

Applications Module

Manage client application onboarding, status, and expiry

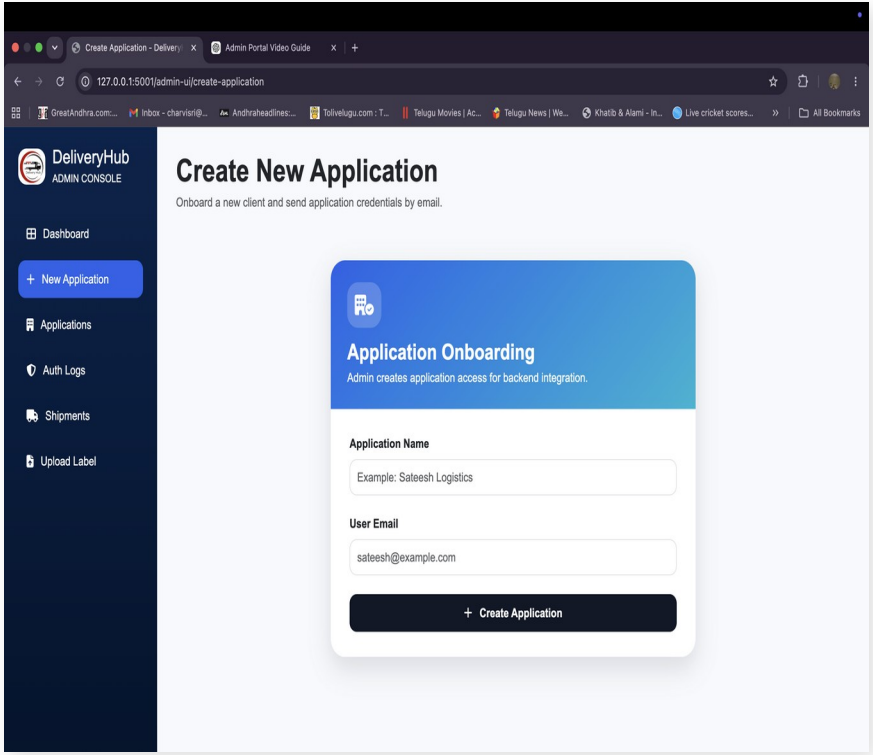


Database Table: applications

`application_id`	UUID — used for auth
`application_token`	Hashed (never plaintext)
`application_name`	Client app name
`user_email`	Credential delivery
`expiry_date`	Validity period
`is_active`	Enable / disable access
`created_at`	Audit timestamp

Create New Application

Onboarding new client applications with secure credential generation

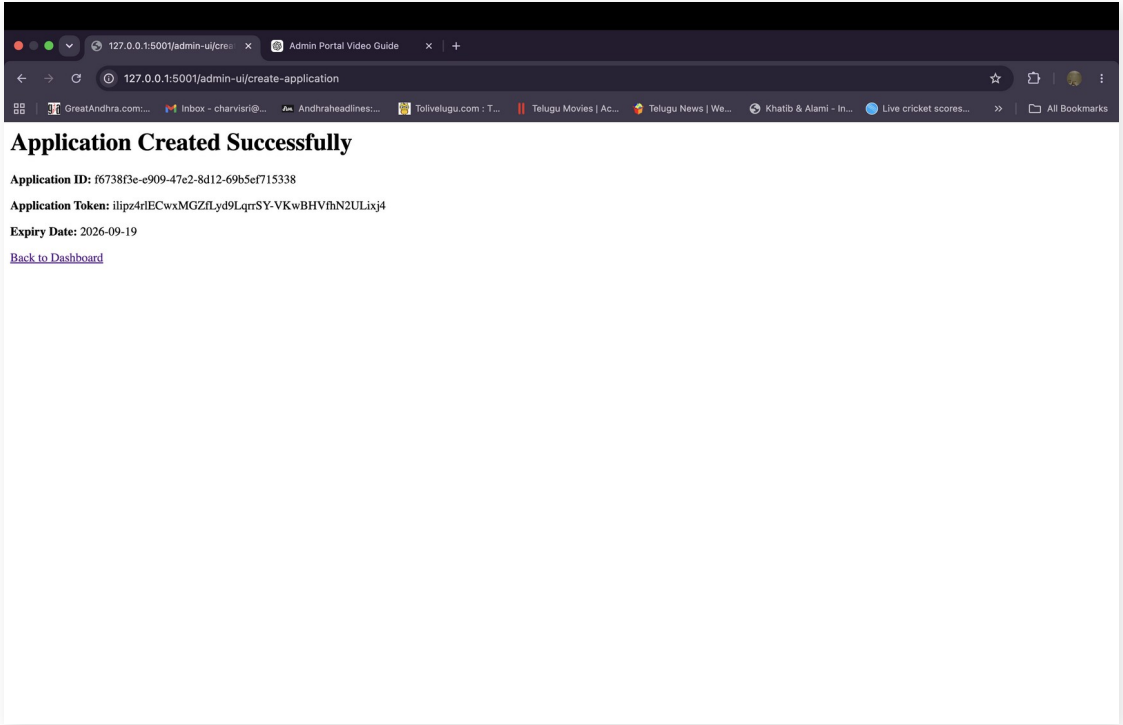


What happens on submit:

1. Generate UUID as Application ID
2. Generate secure random token
3. Hash token before storing
4. Save application to MySQL
5. Set default 90-day expiry
6. Trigger Notification Service

Credential Generation

Token shown once — only its hash is stored in the database



One-Time Display

Token visible only on this screen

Hash-Only Storage

Database never stores plaintext token

Auto Expiry Set

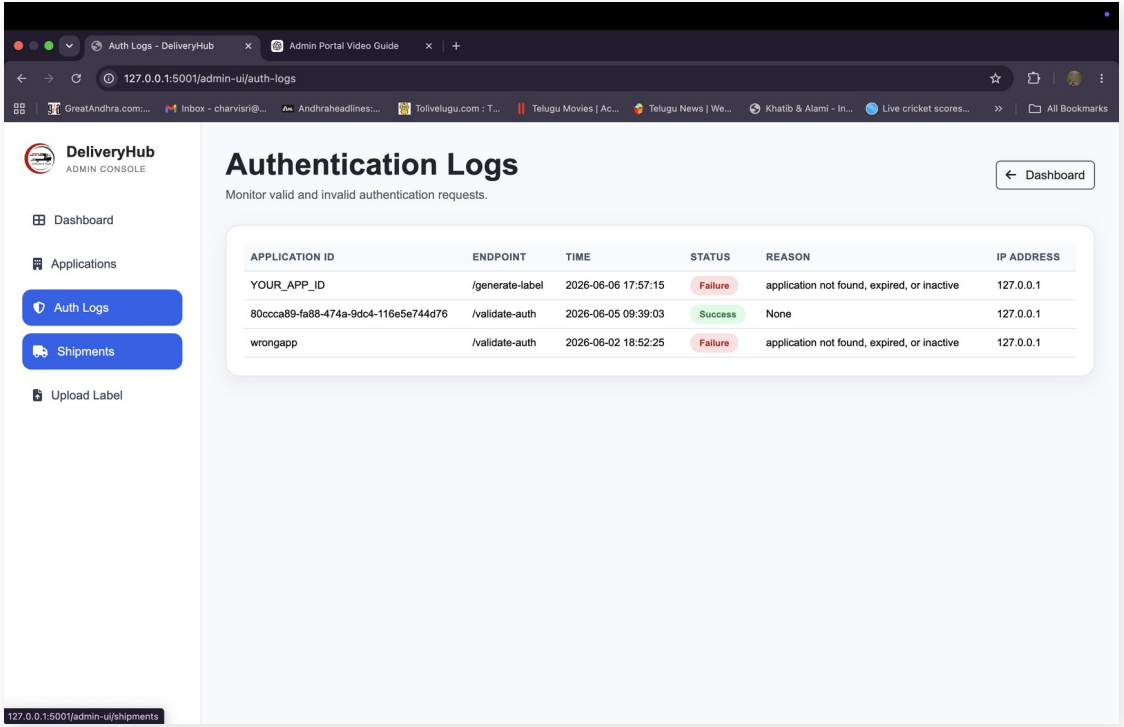
90-day default validity period

Email Delivered

Credentials sent automatically to owner

Notification Service

Automated email delivery of application credentials



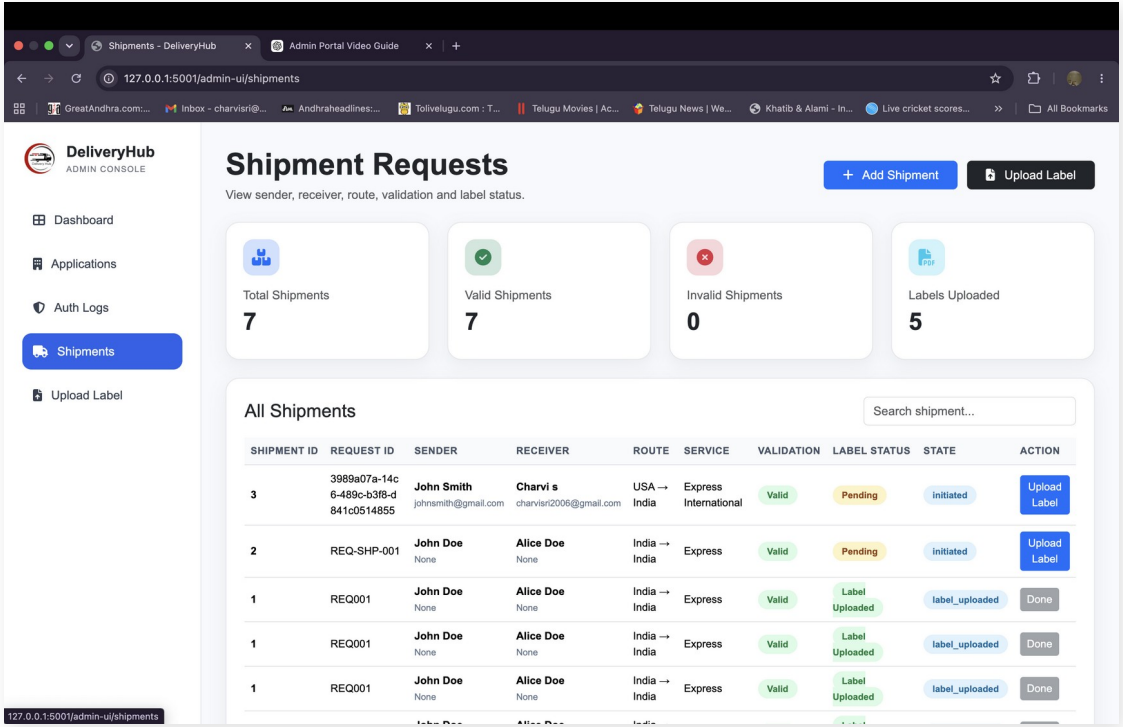
Email Contains:

- ✓ **Application ID**
- ✓ **Application Token**
- ✓ **Expiry Date**

Implemented as a separate notification service integrated with Flask — eliminates manual credential sharing.

Authentication Logs

Full audit trail of every API authentication attempt



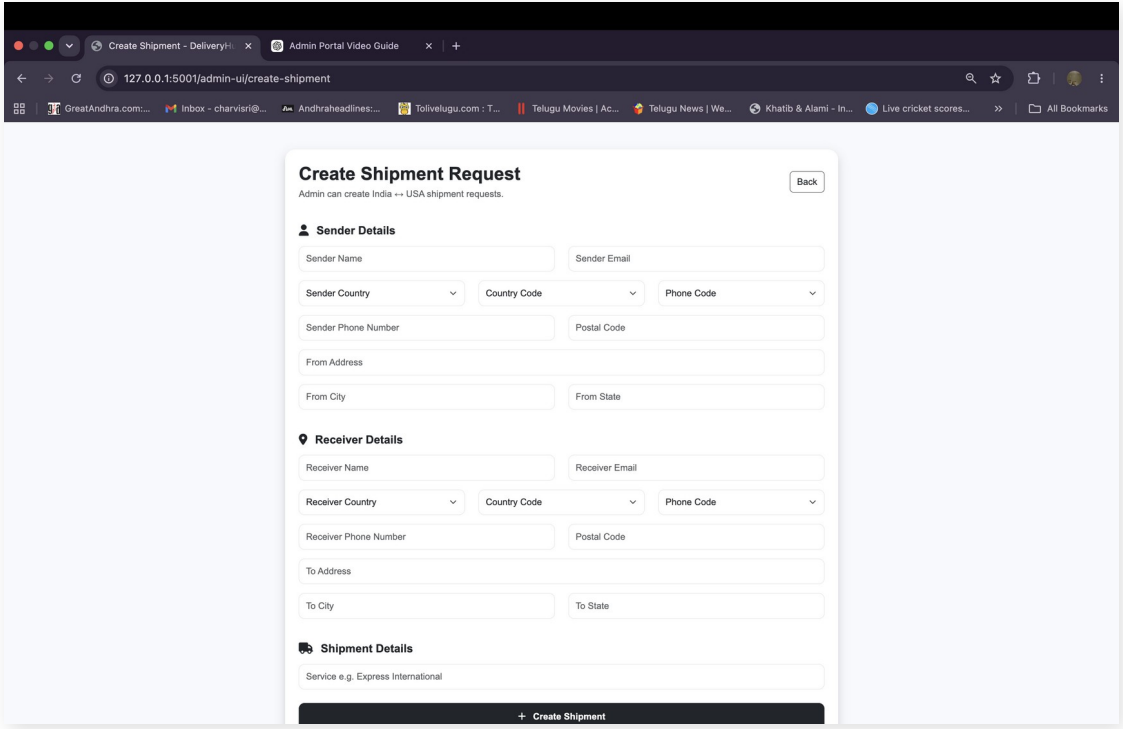
Log Fields

Application ID	Which app made the request
Endpoint	API path accessed
Timestamp	Exact time of attempt
Status	Success or Failure
Reason	Failure explanation
IP Address	Origin of request

Failure Reasons: App not found • Token mismatch • Expired • Inactive

Shipment Requests

Track all shipments, validation status, label status, and lifecycle state



Shipment States

Valid — Passed all validations

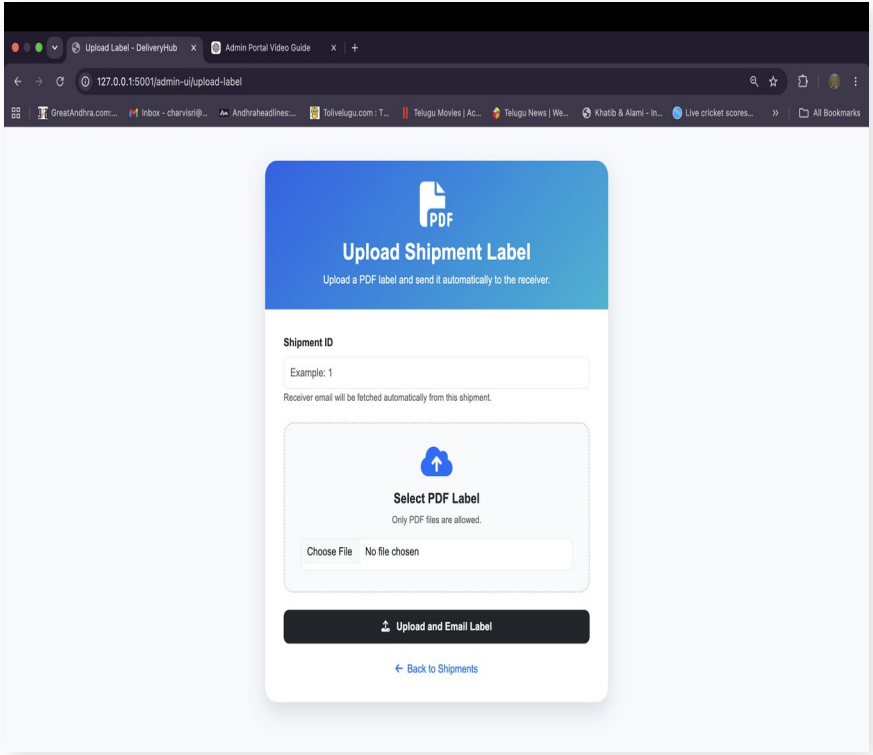
Invalid — Failed validation rules

Pending Label — Awaiting label upload

Completed — Label uploaded & sent

Create Shipment Request

Full sender/receiver form with country validation and DB multi-table write



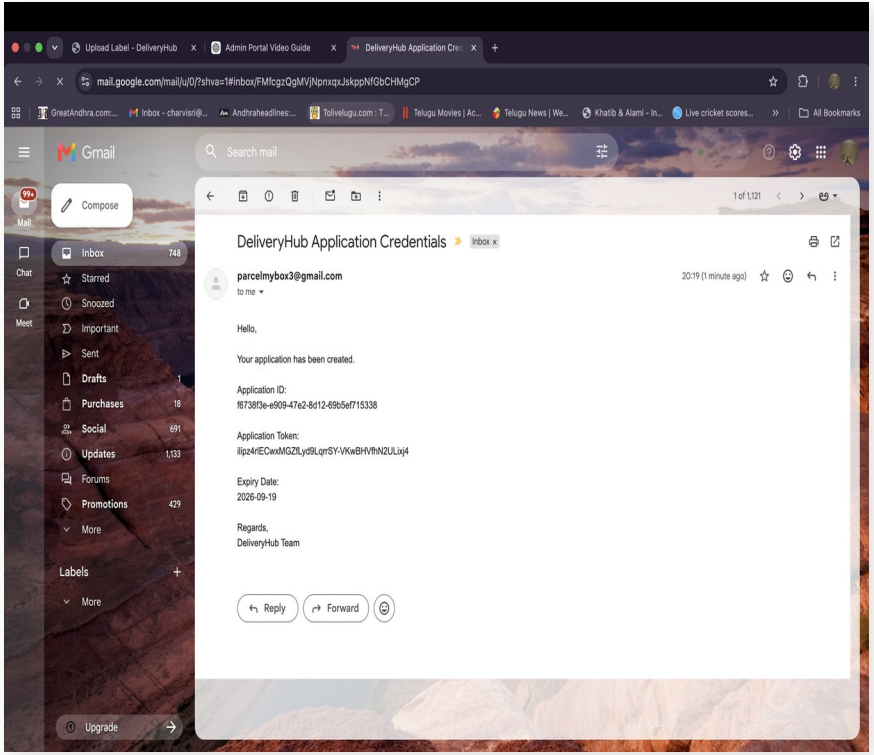
Backend creates records in:

`customers`	Sender & receiver records
`addresses`	From/To address details
`shipments`	Core shipment record
`shipment_tracking`	Tracking entry created

Validates input → country → address → customer → shipment → tracking

Upload Shipment Label

PDF label upload with automatic email delivery to receiver



Upload Process:

1. Check shipment exists in database
2. Validate file has .pdf extension
3. Save file securely on server
4. Update shipment label status
5. Email label PDF to receiver

Security Features

Security was a major focus of this project

Token Hashing

Tokens stored only as bcrypt hashes — plaintext never persists in DB

Expiry Validation

Every auth request checks expiry date — expired apps auto-fail

Active Status Check

Disabled applications cannot access any API endpoint

Authentication Logging

Every attempt recorded with timestamp, reason, and IP address

PDF Validation

Label upload strictly validates .pdf extension before storing

One-Time Token Display

Credentials shown once — never retrievable again after creation

Future Work

Next steps to take DeliveryHub to production

Dockerization

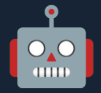
Containerize all services with Docker Compose

Production Deployment

Deploy on cloud infrastructure

Role-Based Access Control

Fine-grained permissions per admin user



Lifecycle Automation

Automate shipment state transitions

Analytics Dashboard

Usage metrics and trend reporting

Microservices

Full containerized microservice deployment

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Thank You

DeliveryHub

Secure application onboarding · API authentication · Shipment management · Label delivery

Built with Flask · MySQL · Python · Email Notification Service